

EECE 571Q

Lecture 12: Magic state distillation, and the true cost of making mistakes

Thursday 18 June 2026

Announcements

- Exam reference sheet due tonight Thursday 18 June 23:59
- Exam on Monday at 15:30-18:00 in HDP 101. Bring student ID and your reference sheet.

Last time

We found the availability of transversal fault-tolerant gate gadgets is code-dependent.

CSS codes : CNOT ✓
H (if $C_1 = C_2$)
S (if weight mult. of 4,
and if $C_1 = C_2$)
T X

$[[15, 1, 3]]$: CNOT ✓
T ✓
H X

Moreover, the Eastin-Knill theorem precludes a single code from having a transversal implementation of a full universal gate set.

Last time

We introduced the $[[15, 1, 3]]$ quantum Reed-Muller code. It has a transversal T and CNOT, but not H .

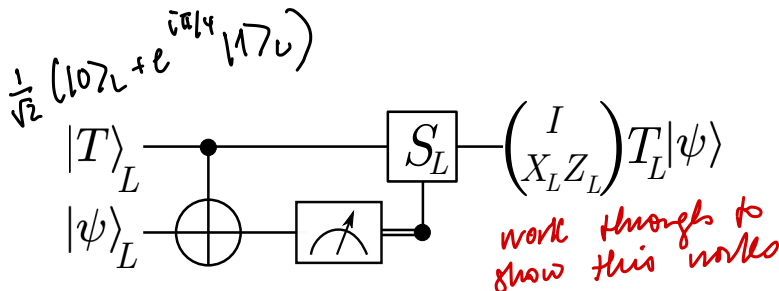
Z	Z	Z	Z	I	I	I	I	I	I	I	I	I	I	I
Z	Z	I	I	Z	Z	I	I	I	I	I	I	I	I	I
Z	I	Z	I	Z	I	Z	I	I	I	I	I	I	I	I
Z	Z	I	I	I	I	I	I	Z	Z	I	I	I	I	I
Z	I	Z	I	I	I	I	I	Z	I	Z	I	I	I	I
Z	I	I	I	Z	I	I	I	Z	I	I	I	Z	I	I
Z	Z	Z	Z	Z	Z	Z	Z	I	I	I	I	I	I	I
Z	Z	Z	Z	I	I	I	I	Z	Z	Z	Z	I	I	I
Z	Z	I	I	Z	Z	I	I	Z	Z	I	I	Z	Z	I
Z	I	Z	I	Z	I	Z	I	Z	I	Z	I	Z	I	Z
X	X	X	X	X	X	X	X	I	I	I	I	I	I	I
X	X	X	X	I	I	I	I	X	X	X	X	I	I	I
X	X	I	I	X	X	I	I	X	X	I	I	X	X	I
X	I	X	I	X	I	X	I	X	I	X	I	X	I	X

Table 11.1: The stabilizer for the 15-qubit code.

Screenshot: Gottesman, Chapter 11

Last time

We briefly introduced the idea of *gate teleportation*, which can be used to apply specific gates given an appropriate resource state.



Logical magic states must be prepared, and this procedure may not be fault-tolerant.

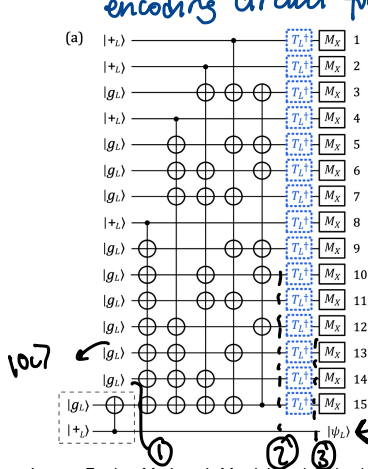
Learning outcomes

- ➊ Perform magic state distillation with the $[[15, 1, 3]]$ quantum Reed-Muller code
- ➋ Outline key assumptions to estimate the physical resources required to implement an algorithm fault-tolerantly
- ➌ Perform a “back-of-the-envelope” resource estimate for an algorithm, and discuss its limitations

Magic state distillation

What if the magic states aren't good enough? *Distill* better ones.

encoding circuit for $[[15,1,3]]$ code.



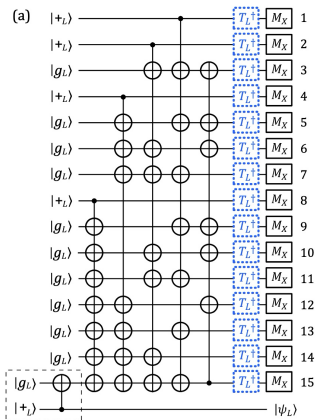
$$\begin{aligned}
 & \textcircled{1} \frac{1}{\sqrt{2}} (|0\rangle|0\rangle + |1\rangle|1\rangle) \\
 & \textcircled{2} \frac{1}{\sqrt{2}} (|0\rangle|0\rangle + |\bar{1}\rangle|\bar{1}\rangle) \quad \text{logical } |0\rangle \text{ in } [[15,1,3]] \\
 & \textcircled{3} \frac{1}{\sqrt{2}} (|0\rangle|0\rangle + e^{i\pi/4} |\bar{1}\rangle|\bar{1}\rangle) \\
 & \textcircled{4} \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} (|0\rangle + |\bar{1}\rangle) |0\rangle + e^{i\pi/4} \cdot \frac{1}{\sqrt{2}} (|0\rangle - |\bar{1}\rangle) |0\rangle \right) \\
 & = \frac{1}{\sqrt{2}} \left[|0\rangle (|0\rangle + e^{i\pi/4} |\bar{1}\rangle) + |\bar{1}\rangle (|0\rangle - e^{i\pi/4} |\bar{1}\rangle) \right] \\
 & \text{Measure } \hat{Z}_L(15) : +1 \text{ w/prob } 1/2 \Rightarrow |\bar{1}\rangle \\
 & \quad \quad \quad -1 \text{ w/prob } 1/2 \Rightarrow |\bar{1}\rangle
 \end{aligned}$$

Image: Fowler, Mariantoni, Martini, and Cleland (2012), *Surface codes: Towards practical large-scale quantum*

computation, Phys. Rev. A 86 (3) 032324.

Magic state distillation

What if the magic states aren't good enough? *Distill* better ones.



Measure and proceed if all $+1$:

$$X_{S1} = X_4 X_5 \cdots X_{10} X_{11}$$

$$X_{S2} = X_1 X_2 X_7 X_{15}$$

$$X_{S3} = X_2 X_5 X_{10} X_{13}$$

$$X_{S4} = X_1 X_2 X_5 X_6 X_9 X_{10} X_{13} X_{14}$$

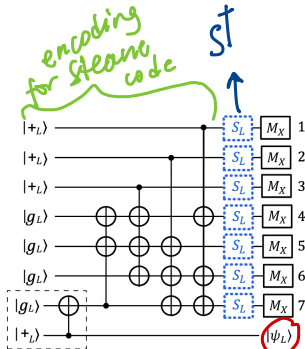
If circuit is perfect and error rate of T is p ,

- success probability $1 - 15p$
- logical error probability $35p^3$

Image: Fowler, Mariantoni, Martini, and Cleland (2012), *Surface codes: Towards practical large-scale quantum computation*, Phys. Rev. A 86 (3) 032324.

Magic state distillation

Easier to understand with this distillation circuit:



if went well,
will be
 $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$

Measure and proceed if all +1:

$$\hat{X}_{S1} = X_1 X_4 X_6 X_7$$

$$\hat{X}_{S2} = X_2 X_5 X_6 X_7$$

$$\hat{X}_{S3} = X_3 X_4 X_5 X_6$$

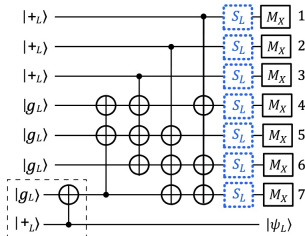
If circuit is perfect and error rate of S is p ,

- success probability $1 - 7p$
- logical error probability $7p^3$

Image: Fowler, Mariantoni, Martini, and Cleland (2012), *Surface codes: Towards practical large-scale quantum computation*, Phys. Rev. A 86 (3) 032324.

Magic state distillation

Easier to understand with this distillation circuit:



$$\begin{array}{l} \hat{X}_{S1} = X_1 X_4 X_6 X_7 + 1 \\ \hat{X}_{S2} = X_2 X_5 X_6 X_7 + 1 \\ \hat{X}_{S3} = X_3 X_4 X_5 X_6 + 1 \end{array}$$

	1	2	3	4	5	6	7
$\hat{X}_{S1}$	1	-1	-1	1	-1	1	1

min wt. logical error is 3.

Possibilities:

235 ←

367

143

165

246

457

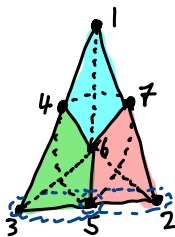


Image: Fowler, Mariantoni, Martini, and Cleland (2012), *Surface codes: Towards practical large-scale quantum computation*, Phys. Rev. A 86 (3) 032324.

$\Rightarrow 7p^3$

↳ Fano plane (2-design)

Magic state distillation

Can perform a similar analysis for the $[[15, 1, 3]]$ code. $35p^3$

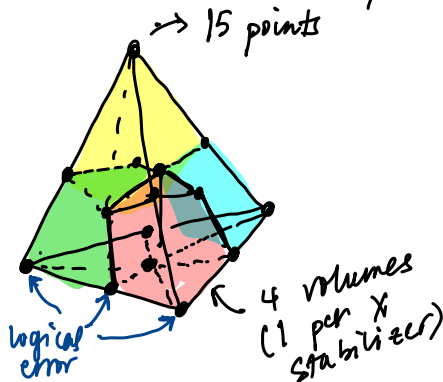
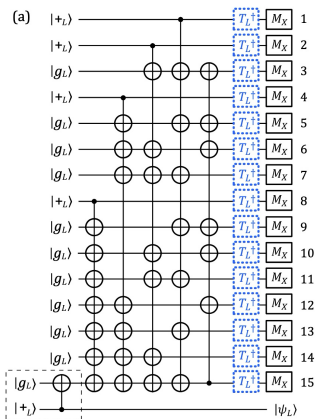


Image: Fowler, Mariantoni, Martini, and Cleland (2012), *Surface codes: Towards practical large-scale quantum computation*, Phys. Rev. A 86 (3) 032324.

Magic state distillation

If error rate isn't good enough, can concatenate:

- first level error probability $35p^3$
- second level error probability $35(35p^3)^3 \sim O(p^9)$
- ...

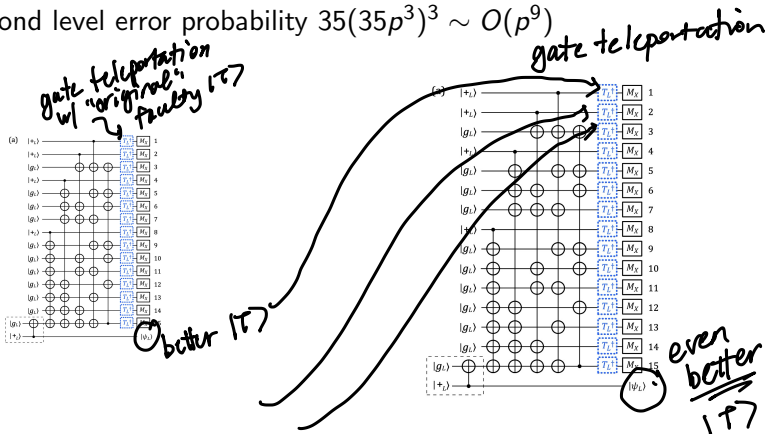


Image: Fowler, Mariantoni, Martini, and Cleland (2012), *Surface codes: Towards practical large-scale quantum computation*, Phys. Rev. A 86 (3) 032324.

☆ See "blank slides"
for content of second
half of lecture
(content did not render
in this PDF)

$$O(2^N) \quad O(\sqrt{2^N})$$

